

The University of Chicago Library houses millions of physical volumes in its on-site facilities, and this collection continues to grow. To manage limited shelf space, the library sought a data-driven approach to identify which books could be moved to off-site storage without disrupting access to users.

The team used machine learning techniques to develop a tool that predicts which books are safe to send to off-site storage. The approach combined ten years of anonymized circulation records with catalog information such as subject headings and call number classifications. With over 95% of books lacking checkout records, relying on observed usage could not meaningfully distinguish between books (see *Figure 1*). Instead, the team grouped books by thematic similarity and estimated demand per-group by transferring insights from frequently borrowed books to similar titles with no checkout history (see *Figure 2*). Each book received a predicted usage score reflecting the borrowing patterns of its broader subject area.

The resulting tool produces ranked storage recommendations for any number of books. This allows library staff to balance space constraints against access needs with evidence-based criteria.

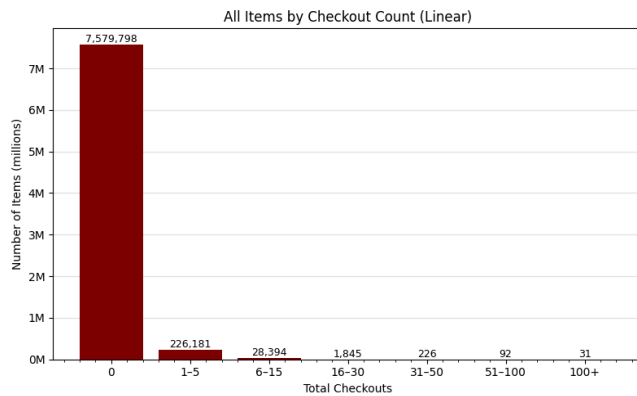


Figure 1: Distribution of checkout data is extremely sparse

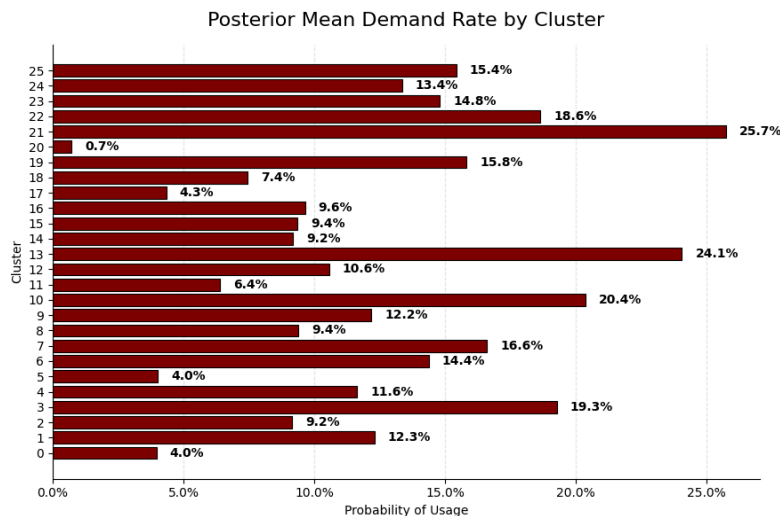


Figure 2: Probability Distribution of each cluster being used and checked out—clusters represent thematically similar books. (e.g., cluster 5 & 21 are dominantly Indo-Iranian languages, then American Literature respectively)